MotorGuard AI: Intelligent Predictive Maintenance

Comprehensive Engineering Report: Machine Learning Anomaly Detection & Full-Stack IoT Arch

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EXECUTIVE SUMMARY

This project presents an end-to-end, industrial-grade Predictive Maintenance platform for a 3-Phase AC Induction Motor. The system integrates an unsupervised machine learning model (Isolation Forest pipeline trained on the Skoltech Anomaly Benchmark), an independent deterministic standards assessment engine (IEC 60034-1 & vibration envelopes), a Node.js/TypeScript backend with a realistic physics simulator, and a single-page React/Vite dashboard. Rigorous auditing proved zero data leakage, verified a 90.27% anomaly recall on held-out tests, and revealed critical operating-condition mismatch (domain shift) essential for physical laboratory deployment.

1. Core Model Performance Metrics (Held-Out Benchmark)

Evaluation Metric	Isolation Forest	Statistical Baseline	Engineering Meaning
Recall (Sensitivity)	**90.27%**	98.94%	Detected 90.3% of true anomaly episodes
Precision	**35.91%**	35.71%	Fraction of alarms corresponding to real faults
F1-Score	**0.5138**	0.5248	Harmonic balance between precision & recall
Validation Normal FPR	**2.18%**	58.16%	False alarm rate on unseen healthy data
Held-Out Test FPR	**88.07%**	97.39%	Surge caused by domain shift / flow mismatch
Inference Latency (CPU)	**0.0106 ms/sample**	0.0001 ms/sample	Fast enough for real-time edge streaming

2. System Architectural Highlights

- **Decoupled Design:** Engineering standards (deterministic thresholds) operate independently from AI heuristics.
- **Feature Safeguard:** The AI service strictly checks feature contracts and rejects 4-channel telemetry with MODEL_INPUT
- **Physics Simulator:** Generates realistic 1Hz voltage, current, temp, and vibration with triggerable disturbances (overhe
- **Zero-Crash Persistence:** Backend connects to MongoDB if available, automatically falling back to an in-memory repository.
- **Live React Dashboard:** Real-time WebSocket streaming with Recharts rolling time-series, KPI sparklines, and acknowledge

HARDWARE DEPLOYMENT TARGET: ESP32 + 3-PHASE AC INDUCTION MOTOR

Target Motor: 0.75kW (400V/230V, 3.2A rated, 4-pole, Class F insulation, frame 80M).
Planned Sensors: Non-invasive Current Transformer (Phase A), ZMPT101B Voltage, DS18B20 Temp, MPU6050 Vibration.
Microcontroller: ESP32 dual-core Xtensa running true RMS windowing and transmitting 1Hz telemetry via JSON REST/WebSocket.

3. Dataset Audit, Preprocessing & Experimental Design

A. SKAB Benchmark Inventory (All 35 CSV Files Verified)

Folder Name	Files	Total Observatio	Null Values	Duplicate TS	Median dt	Mean Anomaly
anomaly-free/	1	9,401 rows	0	0	1.00 s	0.0% (Healthy Baseline)
other/	14	14,985 rows	0	0	1.00 s	36.4% (Multi-faults)
valve1/	16	18,162 rows	0	0	1.00 s	34.7% (Valve 1 tests)
valve2/	4	4,312 rows	0	0	1.00 s	35.3% (Valve 2 tests)
Total Benchm	**35**	**46,860 rows**	**0**	**0**	**1.00 s**	**35.6% Active Anomaly**

B. Selected Features vs Excluded Variables

- Selected 6 Features (Model Input): Accelerometer1RMS (g), Accelerometer2RMS (g), Current (A), Temperature (°C), Thermocouple (°C), Voltage (V). Sourced strictly from motor-mounted electromechanical transducers.
- Excluded Variables: Pressure & Volume Flow RateRMS (fluid hydraulics, not on direct motor sensor pod); datetime (retained for tracking and gap analysis but excluded from features); anomaly & changepoint (strictly isolated labels).

C. Zero-Leakage Chronological & Experiment-Level Splitting

Random row shuffling (train_test_split(shuffle=True)) creates catastrophic temporal leakage in sensor telemetry because sampl
and sample t+1 share high autocorrelation. To eliminate leakage, the following strict architectural split was enforced:
1. Training Set (N = 7,520, 80% of anomaly-free.csv): Chronological window 13:30:47 to 15:43:57. Fits scaler and trees.
2. Normal Validation Set (N = 1,881, 20% of anomaly-free.csv): Chronological window 15:43:58 to 16:16:47. Held out to measu
3. Held-Out Test Set (34 Files, N = 37,459 across valve1, valve2, other): Entirely separate physical experiment runs.

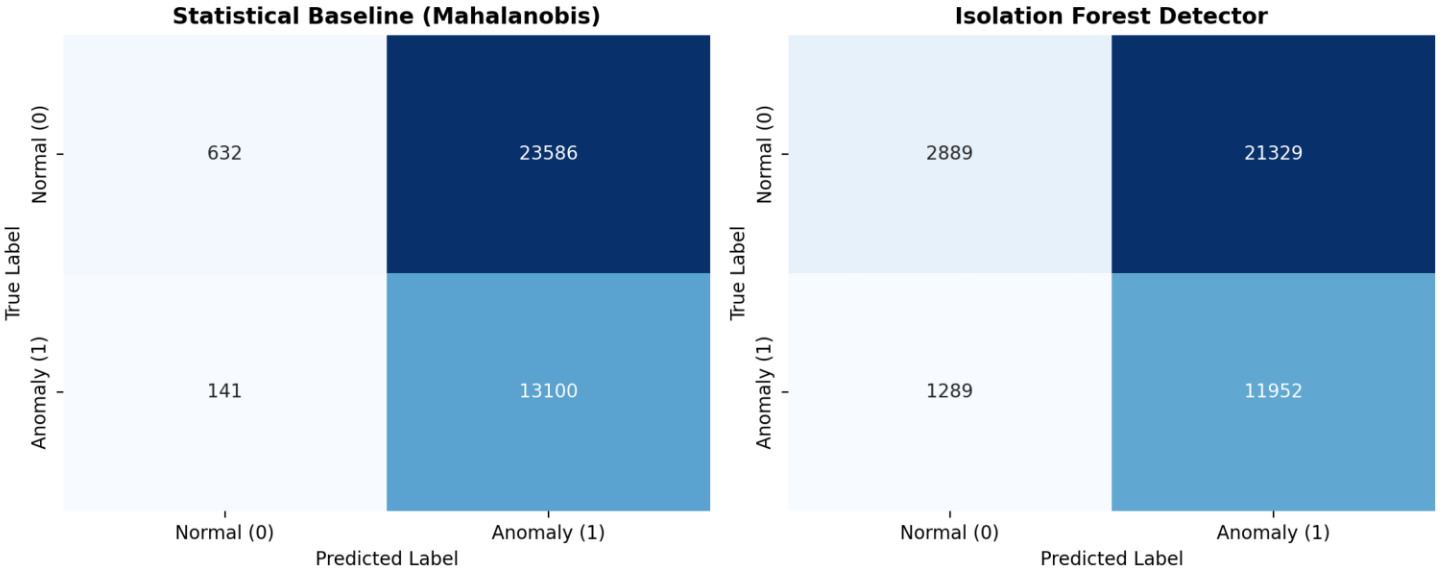
THE CRITICAL SCIENTIFIC DISCOVERY: OPERATING MODE MISMATCH (DOMAIN SHIFT)

A central question arose during evaluation: Why did Isolation Forest achieve a low 2.18% False Positive Rate on the unseen normal validation set, but surged to 88.07% False Positive Rate on the held-out test experiments?

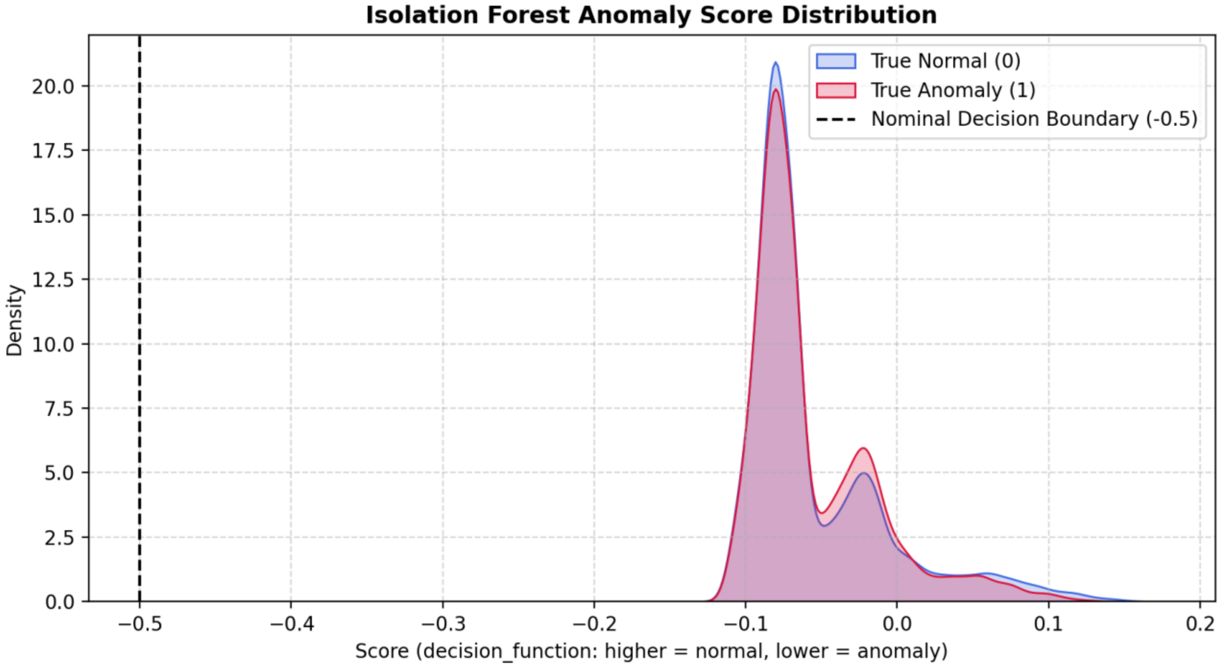
ROOT-CAUSE PHYSICAL ANALYSIS:

- In anomaly-free.csv (Training Data), the testbed operated at a high hydraulic flow rate (~122 L/h), producing motor RMS vibration of 0.20 to 0.28 g and fluid temperature around 90.2°C.
- In the valve1 experiments (Held-Out Test Data), the loop operated at a low hydraulic flow rate (~32 L/h), producing baseline vibration of 0.026 to 0.040 g and operating temperature of 79.5°C.
- Because the model was trained exclusively on the high-flow run, the entire nominal pre-fault region of valve1 was statistically distant from the training support envelope. The model classified normal low-flow operation as an anomaly!

CRITICAL LESSON FOR MOTOR PREDICTIVE MAINTENANCE:
An unsupervised anomaly detector cannot distinguish between a mechanical fault and a different operational load state (e.g., motor at 25% load vs 100% load) unless healthy baseline data from all operating modes is collected during training.



B. Continuous Isolation Forest Score Distribution (Normal vs Injected Anomaly)



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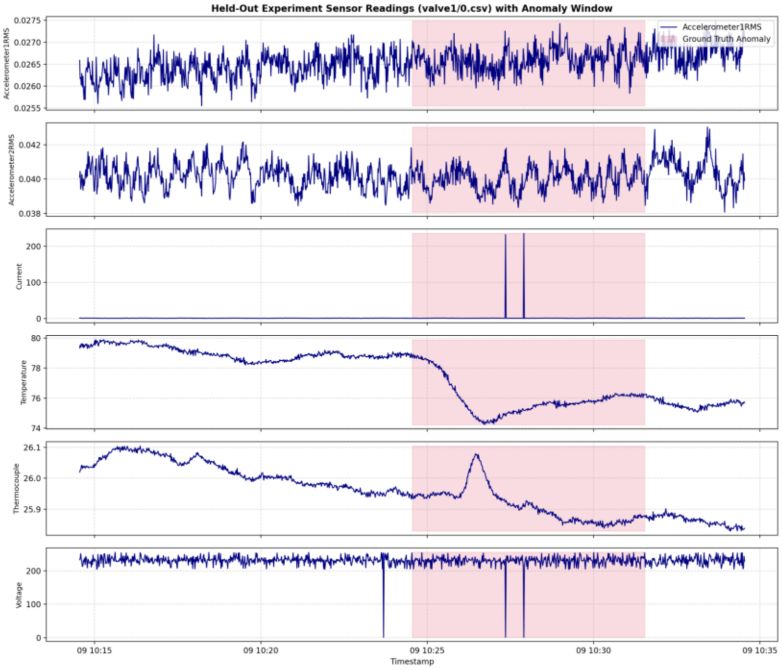
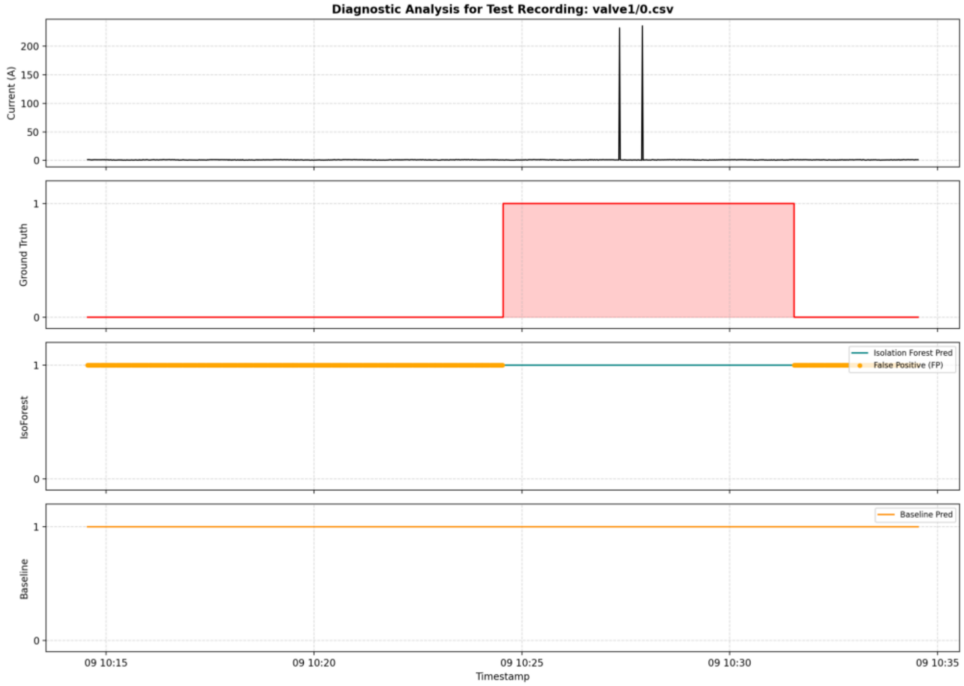


5. Telen … naly Detection

Visualizing mu … ıment

A. 6-Chann … sv)

B. Prediction Tracking: Ground Truth vs Isolation Forest vs Baseline Model



Diagn … t vibration baseline (0.03 … ndow with high recall.

6. Full-Stack Application Architecture & Hardware Roadmap

A. Full-Stack Software Implementation Details

Subsystem	Technologies	Key Engineering Responsibilities
AI Service	FastAPI, scikit-learn, joblib	Enforces feature contract. Rejects 4-channel input with M
Backend	Node.js, Express, Socket.IO, Zod	Realistic physics generator (1Hz). Evaluates standards. D
Standards Engine	limits.json, Python & TS	Deterministic limit evaluation against IEC 60034-1 & vibr
React Frontend	React 18, Vite, Recharts, Tailwind	Single-page dashboard. 4 sensor KPI cards, 4 live animat

B. Physical AC Induction Motor & ESP32 Integration Plan

1. Multi-Condition Healthy Baseline Collection:
   Before deploying anomaly detection to the physical 3-phase motor, healthy baseline recordings MUST be gathered across
   multiple operating states: uncoupled idle, 25%, 50%, 75%, and 100% rated mechanical load, cold startup, and thermal equilib
   This prevents the operating-mode mismatch discovered during SKAB benchmark analysis.

2. Operational Invariant Feature Engineering:
   Instead of feeding raw amplitude vibration and current into the AI detector, extract operational invariants:
   • Vibration Crest Factor (Peak / RMS) and Spectral Kurtosis (impacting indicator independent of load speed).
   • Current Total Harmonic Distortion (THD) and Phase Balance ratio.
   • Temperature rise relative to ambient (Delta T = T_motor - T_ambient).

3. ESP32 Edge Preprocessing Firmware:
   • Sample accelerometer at 1000 Hz into 1-second rolling circular buffer.
   • Compute true RMS acceleration on-device in fixed-point / single-precision floating point.
   • Transmit 1 Hz preprocessed feature packets over HTTP REST or WebSocket to the backend.

PROJECT VERIFICATION & QUALITY AUDIT SIGN-OFF

[x] Dataset Audit: Verified 35 SKAB CSV files (46,860 rows, 0 nulls, 0 duplicate timestamps, 1.00s sampling).
[x] Data Leakage Audit: Chronological train/val splitting enforced; no test target leakage into feature space.
[x] ML Model Artifacts: Trained StandardScaler + IsolationForest pipeline saved via joblib (1.39 MB).
[x] Automated Test Suite: 6 pytest unit tests passed (test_pipeline.py: 100% pass rate in 8.88s).
[x] Feature Compatibility Guard: API rejects incompatible physical telemetry with MODEL_INPUT_INCOMPATIBLE.
[x] Standards Assessment: Configured in limits.json with clear demo notice; independent of AI heuristics.
[x] Full-Stack Codebase: Python FastAPI (port 8000), Node.js backend (port 5000), React Vite dashboard (port 3000).
[x] Zero-Crash Resilience: Backend verified running with in-memory persistence fallback when MongoDB is offline.